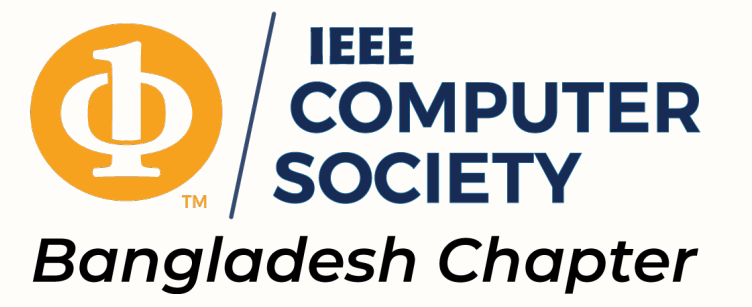


Poster Template for IEEE Computer Society Bangladesh Chapter Symposium 2024



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Introduction

Agricultural productivity is crucial for global food security, with crop health being a key factor in yield. In Bangladesh, the bottle gourd (*Lagenaria siceraria*) is a vital vegetable crop, valued for its nutritional and medicinal benefits. However, its productivity is often threatened by diseases that can severely impact yields and farmers' livelihoods.

Traditional disease detection methods for bottle gourd leaves in Bangladesh rely on manual inspection by experts. While accurate, this approach is labor-intensive, time-consuming, and impractical for large-scale farming. Additionally, the subjective nature of visual assessments can lead to inconsistent diagnoses and improper treatments. This highlights the need for an efficient, reliable, and scalable disease detection and classification method. Recently, Convolutional Neural Networks (CNNs) have emerged as powerful tools for image recognition and classification. CNNs have shown exceptional performance in various fields, such as medical imaging, facial recognition, and autonomous driving, by automatically extracting and learning features from images. This study aims to leverage CNNs to develop an automated disease classification system for bottle gourd leaves in Bangladesh.

The research focuses on creating a CNN-based model to accurately identify and classify diseases affecting bottle gourd leaves using a dataset of labeled images of healthy and diseased leaves. The primary objective is to help Bangladeshi farmers and agricultural practitioners make informed crop management and treatment decisions. Implementing this system can enhance disease diagnosis accuracy, reduce dependency on expert intervention, and improve agricultural efficiency in the region.

In summary, this thesis explores the application of CNNs for classifying diseases in bottle gourd leaves in Bangladesh. By developing an automated, reliable, and scalable detection solution, this research aims to advance precision agriculture and promote sustainable farming practices. Successful implementation could significantly improve the livelihoods of Bangladeshi farmers and ensure the sustainability of bottle gourd cultivation.

Methodology

Convolutional Neural Network (CNN) using the Keras Sequential API, tailored for a three-class image classification task. The model processes input images with dimensions (Batch Size, Image Size, Image Size, Channels) through initial preprocessing steps, including resizing, rescaling, and data augmentation, to enhance the training dataset's variability and robustness.

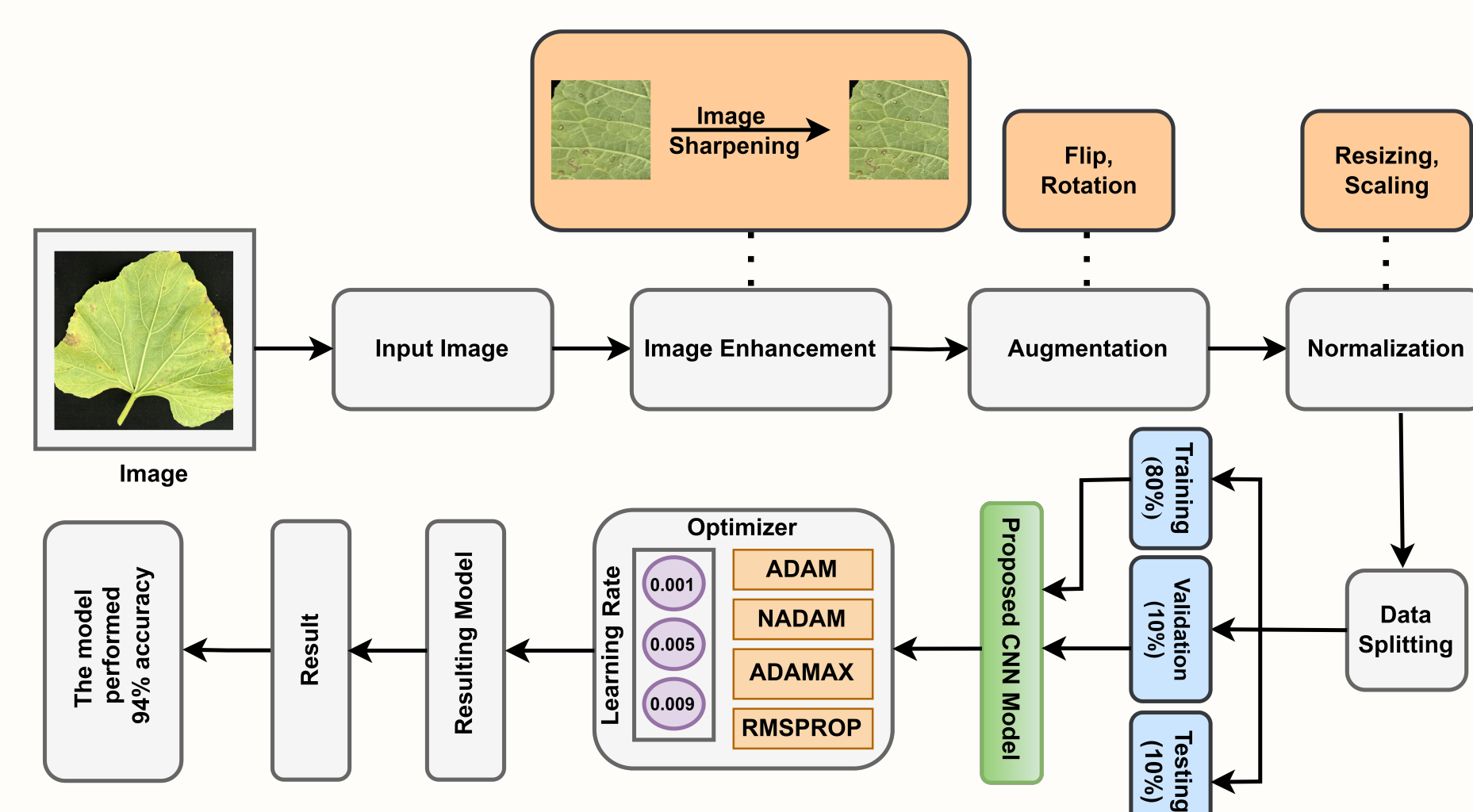


Figure 1: Proposed Methodology.

The network architecture comprises six convolutional layers, each equipped with 32 to 64 filters of size (3,3) and ReLU activation, followed by max-pooling layers to reduce spatial dimensions and computational load while retaining critical features. After the convolutional and pooling stages, the feature maps are flattened and passed through a dense layer with 64 units and ReLU activation to introduce non-linearity and learn complex patterns. The final layer is a dense layer with three units and a Softmax activation function, producing probability distributions over the three classes. The model is compiled with the Adam, Nadam, Adamax and RMSProp optimizer, Sparse Categorical Cross entropy loss function, and accuracy as the evaluation metric, ensuring effective training and performance tracking for this multi-class classification problem.

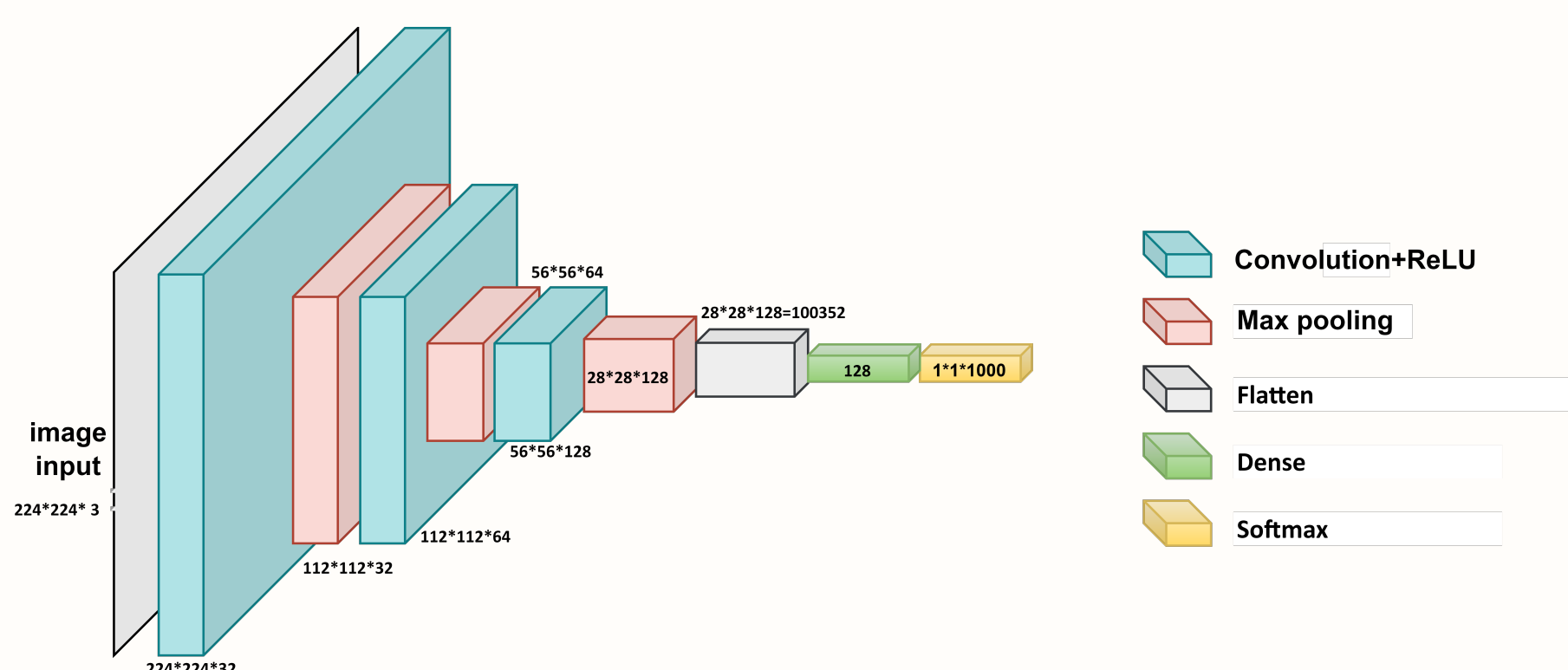


Figure 2: CNN Model Architecture.

Result and discussions

After training our model with above mentioned optimizers for 0.001, 0.005, and 0.009 learning rates, we have taken the model with the best accuracy. Adam optimizer with the learning rate of 0.001 gave us the best model according to the accuracy, loss, and performance analysis.

Table 1: This is a table with training results.

Op	Lr	Tr-Acc	Val-Acc	Tr-Loss	Val-Loss
Adam	0.001	0.312	92.334	0.245	93.897
	0.005	0.556	91.704	0.256	92.051
	0.009	0.614	90.390	0.287	91.669
Nadam	0.001	0.375	92.050	0.257	93.557
	0.005	0.529	90.210	0.357	91.887
	0.009	0.879	85.870	0.815	90.223
Adamax	0.001	0.151	92.575	0.198	93.887
	0.005	0.250	90.160	0.236	92.982
	0.009	0.315	91.175	0.281	91.334
RMSProp	0.001	0.477	92.390	0.259	95.663
	0.005	0.556	88.540	0.335	94.664
	0.009	0.401	91.660	0.5138	92.443

The model trained with the proposed model is good enough to detect the diseases mentioned earlier. It can further be developed with better accuracy if a better and larger dataset is provided.

Future Scopes

The future prospects for bottle gourd leaf disease detection with Convolutional Neural Networks (CNNs) hold immense promise, with opportunities for innovative advancements on multiple fronts. One avenue for exploration involves the development of mobile applications equipped

with CNN-based disease detection models, enabling farmers to conduct real-time monitoring and diagnosis of bottle gourd leaf diseases directly in the field. Furthermore, the development of a comprehensive model capable of detecting multiple diseases simultaneously will enhance its utility, providing a versatile tool that can distinguish between various afflictions affecting the crop. Such advancements would not only increase crop yield and quality but also contribute to sustainable agricultural practices by enabling early detection and reducing the need for broad-spectrum pesticides. As CNN technology continues to evolve, integrating these solutions into a user-friendly mobile platform will revolutionize disease management in agriculture.

Summary and conclusions

In conclusion, the utilization of Convolutional Neural Networks (CNNs) for bottle gourd leaf disease detection represents a pivotal advancement in agricultural technology. By harnessing CNNs, researchers and farmers gain a powerful tool for accurately identifying and classifying diseases affecting bottle gourd crops, enabling timely intervention and mitigation measures. This technology holds significant economic value by reducing yield losses, optimizing resource allocation, and minimizing the need for costly chemical treatments, thereby enhancing overall crop productivity and profitability. Furthermore, the scalability and adaptability of CNN models make them well-suited for widespread deployment across diverse agricultural landscapes, promising to revolutionize disease management practices on a global scale. Looking ahead, the future of bottle gourd leaf disease detection with CNNs holds promise for further innovation and refinement, facilitating the development of more efficient, sustainable, and resilient farming practices that ensure food security and economic prosperity for generations to come.

References

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